Vol 16 Chapter 9 — Composite Substrate-Mass-Mechanism (gen-2 + HYBRID class)

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Vol 16 Chapter 9: Composite Substrate-Mass-Mechanism

9.0 Position and thesis

Vol 16 Ch 4 (Elie) established the volume's core thesis: every BST observable is a matrix coefficient of a substrate operator on $H^2(D\_IV^5)$. Ch 5 (Bergman kernel algebra) and Ch 6 (Casey #14 restriction sequence) supplied the operator-algebra machinery. Ch 9 applies that machinery to the hardest observables in the program — the charged-lepton and Higgs masses — and shows that they are not one mechanism but three, organized by which region of D_IV^5 the relevant K-type lives in.

The chapter's claim is structural, not numerical: the mass observables partition by substrate region, and that partition is the content. The numbers (m_μ/m_e = 207, m_H = 125.28 GeV) are corollaries.

9.1 The mechanism-class TRIPLE (organizing principle)

The charged-lepton + Higgs mass observables sort into three disjoint classes by K-type substrate region:

Class	Substrate region	K-types	Observables
Mersenne-edge	Shilov-boundary edge K-types	gen-0, gen-3	m_e, m_τ
Bergman-bulk	interior $H^2(D\_IV^5)$ bulk K-types	gen-1, gen-2	m_μ, m_τ/m_μ
HYBRID	central vacuum V_(0,0) + boundary	—	m_H (Higgs)

This is the candidate 7th audit category (Keeper K229, SUBSTRATE-MECHANISM-CLASS-PARTITION). The operator-algebra statement of the partition is: the mass operator does not commute with the bulk/boundary (Hardy) projection, so its spectrum splits into a bulk part, an edge part, and a cross (hybrid) part. Each class is one block of that decomposition.

Why this is more than bookkeeping: if the partition were arbitrary, generations could be reassigned freely. They cannot — gen-1/gen-2 sit in the bulk because their K-types V_(1/2,1/2) and V_(3/2,1/2) have half-integer Casimir and are interior; gen-0/gen-3 sit at the edge. The spin-statistics-like sorting (Elie Toy 3921: half-integer Casimir = bulk/fermionic domain, integer Casimir = edge/Mersenne domain) is the forcing argument and is the chapter's central multi-week gate.

9.2 Mass observable = matrix coefficient

For a charged lepton of generation g sitting in K-type V_λ, the mass ratio to the electron is read as a ratio of Bergman matrix elements of the substrate mass operator M:

$$\frac{m_g}{m_e} = \frac{\langle V_{\lambda(g)} \mid M \mid V_{\lambda(g)}\rangle}{\langle V_{\lambda(0)} \mid M \mid V_{\lambda(0)}\rangle}$$

When M is the Bergman kernel operator restricted to a bulk K-type, the matrix element is a Pochhammer ratio (Ch 5, Sec. 5.4): ⟨V_(λ_1,1/2)|K_B|V_(λ_1,1/2)⟩ ∝ c_FK·(n_C/rank)_k/k! for the Sym^k component. The Pochhammer cascade across the spinor tower (Ch 5, Sec. 5.3):

$$(5/2)_1 = 5/2, \qquad (5/2)_3 = 315/8, \qquad (5/2)_5 = 45045/32$$

9.3 Bergman-bulk family: m_μ/m_e (Composite v0.5)

The gen-2 muon (K-type V_(3/2,1/2), Sym^3) gives the Composite v0.5 form:

$$\frac{m_\mu}{m_e} = \underbrace{n_C \cdot (5/2)_3}_{\text{bulk Bergman}} + \underbrace{N_c^4/2^{N_c}}_{\text{edge correction}} = \frac{1575}{8} + \frac{81}{8} = 207 \quad (0.112\%)$$

Forcing status (per F32 L4-5e closure): this is a PARTIAL-FORCING form, Tier 2 STRUCTURAL at the Tier 1 boundary. Three skeleton ingredients are forced — the codim-4 exponent (Casey #14 STANDING), the Bergman matrix-element structure (FK Ch. XII Sec. VI), and the kernel exponent 5/2 = n_C/rank. Five assembly ingredients are form-identified: the additive split, the n_C prefactor, base 2^{N_c}, base N_c^4, and the Pochhammer length 3. The length-3 must be pinned to the Sym^3 degree of the gen-2 K-type and not to N_c=3 (integer-collision flag, Cal #242). The 0.112% is the expected Two-Tier structural floor, not a missing factor.

This is exactly the operator-algebra reading Vol 16 should give: a mass ratio = (bulk matrix element) + (edge matrix element), with the "+" demanding a forcing argument (Sec. 9.6).

9.4 Mersenne-edge family: m_e, m_τ

The gen-0 and gen-3 leptons live at the Shilov edge, where the relevant Schur scalar is the Mersenne ratio 8/7 = 2^{N_c}/M(N_c) (Ch 7 / Elie SSG-8, integer-Casimir/edge domain). The cross-family ratio m_τ/m_μ = 3479/207 ≈ 16.81 at 0.06% reads as a Mersenne-edge (gen-3) / Bergman-bulk (gen-2) quotient — i.e. it crosses two of the three classes, which is why it is a cleaner observable than either mass alone (the shared electron-anchor cancels). The tau itself (gen-3) is edge; the muon (gen-2) is bulk; the ratio is the inter-class matrix-coefficient quotient.

9.5 HYBRID class: the Higgs mass

The Higgs is neither a bulk nor an edge K-type observable — it is the central vacuum V_(0,0) coupled to the boundary, hence HYBRID. The forward form:

$$m_H = \pi \cdot n_C \cdot m_p \cdot \frac{N_{max} - 1}{2^{N_c+1}} = \pi \cdot 5 \cdot m_p \cdot \frac{136}{16}$$

Numerically (m_p = 938.272 MeV): m_H = 125.276 GeV.

Honest precision pin (correction to Friday F19): against the current PDG central value m_H = 125.25 GeV this is 0.021%, not the 0.14% quoted Friday — that figure was computed against an older central value (125.1 GeV). The prediction is better than previously stated; the chapter must quote against the live PDG number and carry the date of the value used.

Mechanism reading: π·n_C is the Bergman bulk-volume normalization (κ_Bergman = −n_C, Ch 8), m_p sets the hadronic scale, (N_max−1)/2^{N_c+1} is the boundary/Mersenne edge factor. The product of a bulk normalization with an edge factor is the operator-algebra signature of the HYBRID class — and it is the same bulk⊗edge structure as the muon's additive split, here multiplicative because the vacuum K-type V_(0,0) has trivial Casimir (the bulk piece is a normalization, not an additive matrix element).

9.6 The bulk⊕Shilov structure (the unifying statement, and the gate)

The chapter's unifying claim: all three classes are blocks of the Hardy decomposition of the mass operator,

$$L^2(D^5_{IV}) = H^2 \oplus H^2_-, \qquad \partial_S D^5_{IV} = S^4 \times S^1/\mathbb{Z}_2,$$

with the mass operator failing to commute with the bulk projection P. Then ⟨V|M|V⟩ = ⟨V|PMP|V⟩ + ⟨V|(1−P)M(1−P)|V⟩ + cross. The bulk block gives the Pochhammer (bulk-family) terms; the (1−P) block gives the Mersenne (edge-family) terms; the cross block gives HYBRID.

This is Gate A from the F32 L4-5e closure, promoted to the chapter's central theorem-to-prove. Concretely: compute the Shilov-boundary matrix element of the gen-2 operator and show it equals $N_c^4/2^{N_c} = 81/8$. If it does, the muon's additive "+" is forced, not fitted — and the same partition that produces the muon edge-term produces the P9 vacuum-subtraction factor 2.02. Same operator decomposition, two observables (lepton mass + cosmological vacuum). That cross-link is a falsifiable Cal #36 Schur-generator candidate for Grace G14.

9.7 Honest tier table

Observable	Class	Form	Precision	Tier	Forced / Identified
m_μ/m_e	Bergman-bulk	1575/8 + 81/8 = 207	0.112%	T2 STRUCT @ T1 boundary	partial (3 forced / 5 identified)
m_τ/m_μ	edge/bulk cross	3479/207	0.06%	T2 STRUCT	cross-class quotient
m_H	HYBRID	$\pi \cdot n_C \cdot m_p \cdot (N_{max}-1)/2^{N_c+1}$	0.021% (vs PDG 125.25)	T2 STRUCT	bulk-norm × edge factor

No observable in this chapter is RIGOROUS-forced yet. The honest claim is the partition (the mechanism-class TRIPLE), which is architectural and testable via Gate A.

9.8 Cross-references

- Ch 5 (Lyra): Bergman kernel algebra, Pochhammer cascade, Hardy decomposition — supplies Secs. 9.2–9.3, 9.6
- Ch 6 (Lyra): Casey #14 restriction sequence — supplies the codim-4 exponent (Sec. 9.3)
- Ch 7 (Lyra + Elie): Bergman-as-matrix-coefficient-sum + SSG-8 Mersenne 8/7 — supplies the edge-family (Sec. 9.4)
- Ch 8 (Keeper + Lyra): $\varkappa$_Bergman = −n_C — supplies the m_H bulk normalization (Sec. 9.5)
- Ch 2 (Lyra + Grace): K-type spectral decomposition — supplies the Sym-degree pin (Gate B, Sec. 9.3)

9.9 Multi-week gates (Cal #189)

- Gate A (additive split): Shilov-boundary matrix element of gen-2 operator = 81/8. Forces the muon "+"; cross-links to P9 factor 2.02. *Highest value.*
- Gate B (Sym-degree pin): gen-2 K-type Sym-degree = 3 from generation index alone, not N_c (Cal #242). Joint Grace Ch 2.
- Gate C (partition forcing): half-integer Casimir → bulk, integer Casimir → edge (Elie Toy 3921) promoted from observation to derivation. This is the K229 7th-category load-bearing step.
- Gate D (HYBRID multiplicative vs additive): why V_(0,0) trivial Casimir makes the Higgs bulk-piece a normalization (multiplicative) rather than an additive matrix element.

9.10 Closure v0.1

Vol 16 Ch 9 scaffolded, Lyra primary. The chapter's content is the mechanism-class TRIPLE as an operator-algebra partition (Hardy blocks of the non-commuting mass operator), with m_μ/m_e (bulk), m_τ/m_μ (cross), and m_H (HYBRID) as the three worked observables. Forcing status is honest throughout: the partition is the architectural claim; the individual numbers are partial-forcing Tier 2 STRUCTURAL; four sharply-scoped gates (A–D) gate RIGOROUS promotion, with Gate A (bulk⊕Shilov additive split, cross-linking the muon edge-term to the P9 vacuum factor 2.02) the highest-value next step.

Carries forward: F32 L4-5e forcing classification (Sec. 9.3); corrected m_H precision 0.021% vs PDG 125.25 (Sec. 9.5).

Tier: Vol 16 Ch 9 scaffolding v0.1; Gates A–D multi-week per Cal #189.

— Lyra, Sat 2026-06-06 10:45 EDT. Vol 16 Ch 9 v0.1 Composite Substrate-Mass-Mechanism (Lyra primary per canonical map): mechanism-class TRIPLE as Hardy-block operator partition; m_μ/m_e = 207 bulk (partial-forcing per F32), m_τ/m_μ edge/bulk cross-quotient, m_H = 125.276 GeV HYBRID (corrected to 0.021% vs PDG 125.25, was 0.14% vs older value); Gate A (Shilov boundary matrix element = 81/8) forces the muon additive split and cross-links to P9 vacuum factor 2.02; Gates A–D multi-week per Cal #189. NOTE: Elie's existing Chapter9 file (CP/Mersenne/Cognition) is on old numbering and needs renumber to a TBD slot — flagged in MESSAGES.